Fiscal Year:	FY 2006	**Task Last Updated:**	FY 02/06/2006
PI Name:	Wilson, Wilbur W. Ph.D.		
Project Title:	High Throughput Screening of Protein-Protein Interactions by microchip SPACE		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bilwil@ra.msstate.edu	**Fax:**	FY 662-325-1618
PI Organization Type:	UNIVERSITY	**Phone:**	662-325-3584
Organization Name:	Mississippi State University		
PI Address 1:	Mississippi State University, Department of Chemistry		
PI Address 2:	Box 9573		
PI Web Page:			
City:	Mississippi State	**State:**	MS
Zip Code:	39762-5613	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	09/01/2001	**End Date:**	11/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Pusey, Marc	**Contact Phone:**	256 544-7823
Contact Email:	marc.pusey@msfc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Henry, C. (Mississippi State University)		
Grant/Contract No.:	NAG8-1837		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The interactions between a mobile phase protein and the same stationary phase protein will be investigated using capillary chromatography. Retention times will be correlated with the osmotic second virial coefficient for the protein in a particular solution (mobile phase) condition. The approach will ultimately be translated to a microchip for true nanoscale analysis. The objective of this work is to utilize nanotechnology to develop a fast screening method for protein-protein interactions related to protein crystallization and protein stability. The development of a nanoscale analysis for protein-protein interactions will allow rapid testing of favorable crystallizing solution conditions as predicted by the crystallization slot. This analysis will impact proteomics by providing a rational approach for robotic designs for screening the many proteins and solution conditions prior to crystallization using high throughput technologies.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	none
Task Progress:	**DRAFT** The SIC methodology was successfully demonstrated as an effective tool for measuring the second virial coefficient for several proteins in a variety of solvent conditions. The accuracy and reliability of the method was verified by performing static light scattering measurements on the same proteins and solvent conditions. The method has been improved to include the capability to perform measurements as a function of temperature.
Bibliography Type:	Description: (Last Updated: 03/13/2008)
Articles in Other Journals or Periodicals	Henry C, Payne R, Valente J, Wilson W, Manning M. "Self-Interaction Chromatography for Screening." Genetic Engineering News 2005;25: 62-63. , Sep-2005
Articles in Peer-reviewed Journals	Valente JJ, Verma KS, Manning MC, Wilson WW, Henry CS. "Second virial coefficient studies of cosolvent-induced protein self-interaction." Biophys J. 2005 Dec;89(6):4211-8. PMID: 16199499 , Dec-2005
Patents	Patent no. 6,974,678. awarded December 2005. Dec-2005 Garcia C, Henry C, WilsonW. "A Micro-fluidic Device for Monitoring Biomolecular Interactions."